



CVE-2025-0141: PALO ALTO GLOBALPROTECT APP PRIVILEGE ESCALATION (PE) VULNERABILITY

Vairav CVE Report

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Vairav Cyber Threat Intelligence Team

Vairav Technology Security Pvt. Ltd.

Thirbam Sadak 148

Baluwatar, Kathmandu

Phone: +977 4541540

Mobile: +977-9820105900

Email: sales@vairavtech.com

EXECUTIVE SUMMARY

A single vulnerability (**CVE-2025-0141**) has been identified in the Palo Alto Networks GlobalProtect App. This flaw enables a locally authenticated, non-administrative user to **escalate privileges to root** on macOS/Linux or NT AUTHORITY\SYSTEM on Windows. With a CVSS v3 score of **8.4 (High)**, this improper privilege assignment can result in complete system compromise if locally exploited.

VULNERABILITY DETAILS

CVE-2025-0141

- **Description:** An incorrect privilege assignment in Palo Alto Networks GlobalProtect App allows a logged in non-administrative user to escalate privileges to root/System by exploiting insecure configuration or file ownership on macOS, Linux, and Windows
- **Impact:** Attackers with basic system access could gain full control—install malware, modify critical services/configuration, access sensitive data.
- **CVSS Score:** 8.4 (High)

AFFECTED VERSIONS

- **macOS & Windows**
 - Versions **< 6.3.3-h1 (6.3.3-c650)** of GlobalProtect App 6.3.
 - Versions **< 6.2.8-h2 (6.2.8-c243)** of GlobalProtect App 6.2.
 - All versions of GlobalProtect App 6.1 and 6.0
- **Linux**
 - Versions **< 6.2.8**

EXPLOIT DETAILS

This vulnerability is particularly severe in shared environments (e.g., enterprise macs, corporate Linux user workstations, or locked-down Windows devices) where non-admin users exist alongside restricted sensitive resources. A malicious or compromised local user

(with login credentials but no admin rights) can launch the exploit which can lead to full system takeover allowing threat actors to install rootkits, exfiltrate data, or disrupt services.

RECOMMENDED ACTIONS

Patch & Upgrade

- **macOS/Windows:**
 - **GlobalProtect App Version 6.3:** Upgrade to Version 6.3.3-h1 (6.3.3-c650) or later.
 - **GlobalProtect App Version 6.2:** Upgrade to Version 6.2.8-h2 (6.2.8-c243) or later.
 - **GlobalProtect App version 6.0 or 6.1:** Upgrade to 6.2.8-h2 (6.2.8-c243) or 6.3.3-h1 (6.3.3-c650) or later.
- **Linux:** Upgrade to Version 6.2.8 or later.

ADDITIONAL SECURITY MEASURES

- **Least-privilege enforcement:** Limit POSIX/Windows user logins only to necessary personnel.
- **File system hygiene:** Use integrity-monitoring tools (e.g., Tripwire, OSQuery, AIDE) to detect unwarranted permission/backend changes.
- **Process auditing:** Enable advanced logging (sysmon for Windows, auditd for Linux/macOS) to capture suspicious privilege escalations.
- **Local host IDS/EPP:** Deploy host-level defense capable of detecting unusual privilege escalation behavior.

REFERENCES

- <https://security.paloaltonetworks.com/CVE-2025-0141>
- <https://cybersecuritynews.com/palo-alto-networks-globalprotect-vulnerability/>

CONTACT US

Vairav Technology Security Pvt. Ltd.

Cyber Defender from the land of Gurkha

Thirbam Sadak 148, Baluwatar

Kathmandu, Nepal

Phone: +977-01-4541540

Mobile: +977-9820105900

Email: sales@vairavtech.com

Website: <https://vairavtech.com>